

When Reading an Equivariance Monitor’s Residual Through the Model Helps, and When It Hurts: Three Sealed Preregistrations and No Forecaster

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Abstract

An equivariance monitor grades a paraphrase-invariance residual $e = h_\ell(gx) - \hat{\rho}_\ell h_\ell(x)$ in raw L2 norm. We ask whether grading it instead in the downstream model’s own read, measured as the model’s output response when e is patched into layer ℓ , catches more real behavioral changes at matched flag rates. The effect is real and it is double-signed. Read through the model, the residual cuts monitor false-clears by 5 to 20 points under backtranslation and 12 to 16 under synonym substitution, replicated at two model scales, but under paraphrase it inverts, adding 35 to 57 points. This raises the practical question the monitor designer actually faces: given a transform, will the consumer read help or hurt. We preregistered three sealed families to answer it, and all three failed their own bars, which we report as executed. No gate predicted the sign: not the absence of a gate (family V1), not the equivariance map’s own held-out fit (family V2), and not transform locality, the held-out R^2 of the identity map $h_\ell(gx) \approx h_\ell(x)$ (family V3), which a clean separation on the earlier families had suggested. The V3 grade is decisive against locality: half-truncation, the most non-local transform, posts the largest advantage in its family rather than the predicted none, and interior word-shuffling, which is representationally local, loses the advantage; the correlation between locality and advantage is -0.05 . The phenomenon replicates, but no scalar measured on the representation forecasts its sign. The honest design consequence is conservative: a consumer-metric gate cannot be certified from a calibration-time representation statistic, and a monitor should keep the raw metric unless the consumer read has been validated empirically on the transform family it will face.

1 Introduction

Internal-equivariance monitors gate on $h_\ell(gx) \approx \hat{\rho}_\ell(g) h_\ell(x)$: a paraphrase g should move a layer’s representation the way a fitted linear map $\hat{\rho}_\ell$ predicts, and a residual that violates this flags a possible spec-gaming or drift [9]. The residual is scored in raw L2 norm. Observation Theory raises an objection that applies to any such monitor [7, 8]. The layer’s activations have a consumer, the downstream computation, and the norm that matters is the consumer’s read of the residual, not its raw magnitude. A large residual in a direction the downstream network ignores is a false alarm; a small residual exactly where the network reads is a false clear, which is the case the monitor exists to catch.

This paper measures the consumer-read alternative. We grade the residual by its measured patch-response: inject e at layer ℓ and record the model’s output-KL response, the consumer’s

actual sensitivity to that residual. The alternative is not uniformly better or worse. It cuts false-clears sharply under some transforms and inverts under others, so the useful question is not whether to read through the model but when. We attacked that question the way we would want a safety claim attacked: three preregistered families, each naming its cells, pass/fail thresholds, and seeds in version control before any evaluation data was opened, verdicts committed as executed. All three failed. The failures are the result. They rule out, in turn, the three variables a designer would reach for first, and the third rules out a variable that had looked clean on the first two families. We report what survives: a robust, replicated, double-signed effect whose sign we cannot yet predict from the representation, and the conservative rule that honesty about that forces.

2 Related work

Our instrument is activation patching, the causal-tracing technique that measures a component’s effect on the output by replacing its activation and reading the change downstream [1, 3]. We use it not to localize a circuit but as the grading norm of a safety monitor: the model’s output-KL response to the injected residual is the consumer’s read. The choice of patching metric is known to change which components look important [2], which is why we fix the metric, the flag rate, and the bars in each preregistration before grading. Reading a representation through a downstream map is the probing-classifier idea [4]; the consumer read applies it to a monitor’s residual rather than to a labeled concept. The object of study is an internal-equivariance monitor, which gates on a fitted paraphrase-invariance map and scores the residual in raw norm [9]; monitoring and steering a model by reading its activations is an active safety direction [5]. The claim that a representation has no single geometry until a consumer reads it, and the read operator that follows, are developed in the Observation Theory program and shipped in its public artifacts [7, 8]. What is new here is neither patching nor consumer-relative reading but the measurement of a monitor built on both, and specifically the sealed negative: the consumer read’s benefit is real and replicated, yet its sign is not forecastable from the three representation statistics we preregistered. We adopt preregistration in the form argued for language research by van Miltenburg et al. [6], with pass/fail bars and seeds frozen before grading and failed families reported as executed. We do not offer a better monitor. We offer the measurement of when reading a monitor’s residual through the model helps, and the evidence that no representation scalar we tried tells you in advance.

3 Method

Setup. An open causal LM (Qwen2.5, 0.5B and 1.5B) reads Social-Chemistry-101 action texts and their transforms gx . For each layer ℓ the whitepaper’s ridge estimator fits $\hat{\rho}_\ell$ on a calibration split; the residual $e = h_\ell(gx) - \hat{\rho}_\ell h_\ell(x)$ is injected at ℓ by a last-position activation patch and the metric is the model’s output-KL response to that injection. Ground truth is the model’s actual next-token KL between x and gx . A pair is “behavior-changed” when that KL exceeds the calibration 75th percentile. The monitor flags the top quartile by each metric, matching the flag rates, and the reported quantity is the false-clear rate, the fraction of behavior-changed pairs not flagged, with the consumer-read advantage

$$\Delta = fc_{\text{raw}} - fc_{\text{true}}.$$

$\Delta > 0$ means the consumer read clears fewer real changes than raw L2.

Discipline. Each family is a sealed preregistration naming its cells, bars, seeds, and manipulation checks before any evaluation data is opened, with a one-day cooling-off between construction and seal. Graded verdicts are committed exactly as the frozen rules grade them. Sizes were set from

measured memory footprints, not from model weights. All records, seals, per-cell graded JSONs, and post-mortems are in the campaign repository.

4 The effect is real, and its sign flips by transform

Read through the model, the residual is a better monitor under representation-preserving transforms and a worse one under representation-relocating transforms. Under backtranslation, the graded cells cut false-clears by 5 to 20 points at matched flag rates (family V1 cell B, -14 and -15 points at 1.5B; family V2 cells D, E, G, from -5 to -20 points across two scales). Under synonym substitution, family V3 cells H and K cut them by 12 to 16 points at both scales. The same read inverts under paraphrase: family V1 cells A and C and family V2 cell F all add 35 to 57 points of false-clears over raw L2, at both model scales. Injecting a residual that is dominated by map error probes the model’s sensitivity to a direction that is not the perturbation, while the raw norm degrades gracefully.

Two findings recur under every seal and we state them as robust. The paraphrase inversion is confirmed at both scales three times over. The equivariance map’s held-out R^2 collapses at the final layer in seven of eight graded runs, from above 0.8 in calibration to at or below zero, so a monitor calibrated at the final layer grades against a broken map regardless of which norm it uses.

5 Two sealed attempts to gate the effect on map fit both fail

Family V1 used no gate. It graded FAIL because its bar required a consumer advantage on every cell and the paraphrase cell inverted. Its post-mortem found the advantage’s magnitude unforecastable from the single-split headroom it had qualified on.

The natural repair is to gate on the equivariance map’s own quality: read through the model only where $\hat{\rho}$ generalizes. Family V2 tested this, gating on the held-out R^2 of $\hat{\rho}$, and graded FAIL, six of eight units passing and the failure precise. In the deep layers of the 1.5B backtranslation cells the map’s held-out R^2 is slightly negative, so the $\hat{\rho}$ -fit gate calls them out-of-regime, yet the consumer advantage persists there at 9 and 17 points. The advantage does not need a good map. It needs the patched residual to point roughly along the true perturbation, and map fit is a leaky proxy for that.

6 The locality law, preregistered and refuted

On families V1 and V2 a single variable had separated the outcomes with no overlap: transform locality, how nearly $h_\ell(gx) \approx h_\ell(x)$, measured as the held-out R^2 of the identity map,

$$\Lambda_\ell = 1 - \frac{\sum \|h_\ell(gx) - h_\ell(x)\|^2}{\sum \|h_\ell(gx) - \overline{h_\ell(gx)}\|^2}.$$

It requires no map fit. Every backtranslation unit, including the two the $\hat{\rho}$ -gate got wrong, had Λ between $+0.28$ and $+0.45$; both paraphrase units had Λ near -1.3 and -1.8 . The reading was mechanistic and plausible: where a transform is local, $h_\ell(gx) = h_\ell(x) + \delta$ with small δ , so even a noisy $\hat{\rho}$ leaves the residual approximately equal to the true perturbation and the consumer read stays informative; where it relocates the representation, the residual is map error and the read inverts.

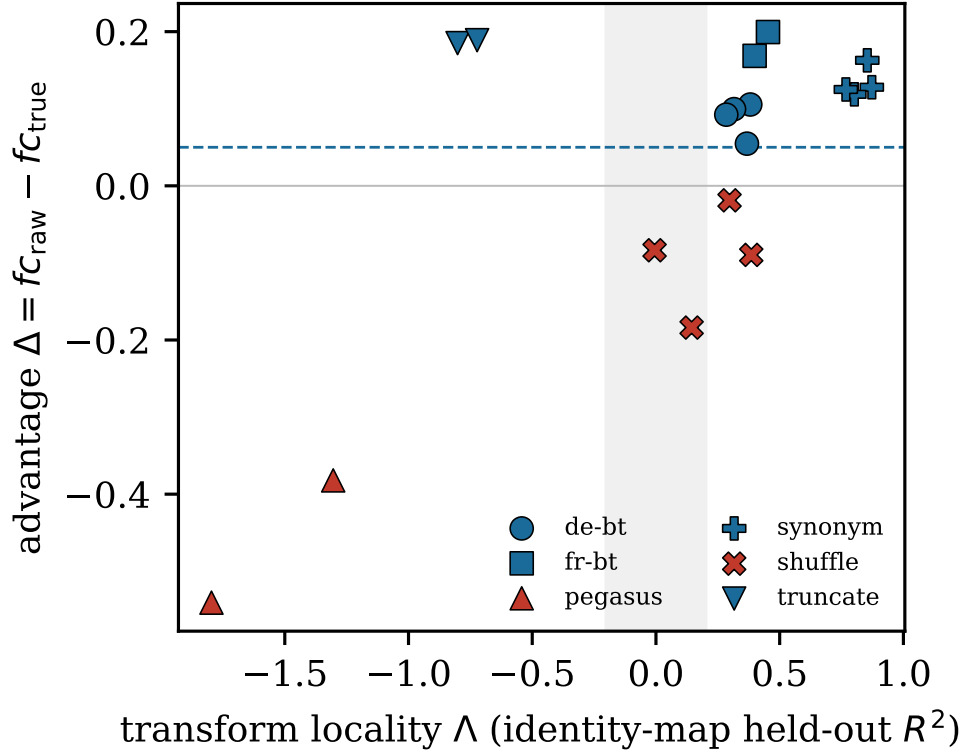


Figure 1: The consumer-read advantage Δ against transform locality Λ , the identity-map held-out R^2 , one point per graded (cell, layer) unit across families V2 and V3. Blue points clear more real changes than raw L2; red points clear fewer. Locality does not order them: the truncation units near $\Lambda = -0.75$ sit as high as the synonym units near $\Lambda = +0.8$, and the shuffle units inside the shaded transition band sit below zero. The earlier families’ clean split was an artifact of the two transforms they happened to contain.

Family V3 preregistered this as a law and tested it on transforms the earlier families did not contain. It graded five cells on a constructed locality ladder, from synonym substitution through interior word-shuffling, which is textually very different yet representationally local, to half-truncation. The sealed predictions were a threshold law, $\Delta \geq 0.05$ when $\Lambda \geq 0.20$ and $\Delta \leq 0.01$ when $\Lambda \leq -0.20$, and a monotonicity, $\text{Spearman}(\Lambda, \Delta) \geq 0.7$ pooled over all units. Both failed, and the failure is not marginal.

Table 1 shows why. Half-truncation, the most non-local transform in the study at $\Lambda \approx -0.75$, does not lose the advantage as the law requires; it posts the largest advantage in the family, about 19 points, where the bar was $\Delta \leq 0.01$. Interior word-shuffling, which is representationally local, loses the advantage instead of keeping it. Pooled over all graded units the correlation between locality and advantage is -0.05 , against a floor of 0.7. The consumer advantage appears at both extremes of locality and vanishes in the middle. The clean separation on families V1 and V2 was a property of the two transforms they happened to contain, not a law.

The V3 grade also corrects the mechanistic reading we drew from V2. Truncation is more non-local than several backtranslation units and does not invert, so “non-local implies inversion” is false. The two transforms that lose the advantage, shuffle and paraphrase, both rearrange or

cell (transform)	scale	Λ	Δ	arm result
H (synonym)	0.5B	+0.87, +0.85	+0.13, +0.16	local: pass
K (synonym)	1.5B	+0.80, +0.77	+0.12, +0.13	local: pass
L (shuffle)	0.5B	+0.39, +0.30	-0.09, -0.02	local: fail
I (shuffle)	1.5B	+0.14, -0.01	-0.18, -0.08	band
J (truncate)	0.5B	-0.80, -0.72	+0.19, +0.19	non-local: fail

Table 1: Family V3, graded (two mid layers per cell). Half-truncation is the most non-local transform yet posts the family’s largest advantage, where the law predicted none; word-shuffling is local yet loses the advantage. Locality does not predict the sign.

rewrite tokens while preserving length and surface plausibility; the two that keep it, backtranslation or synonym and outright truncation, either barely move the tokens or delete them. A natural reading is that what governs the sign is whether the decision-relevant change survives into the patched residual. We tested that reading post-hoc across all graded units, and it does not forecast the sign either. A directional survival measure, the cosine between the patched residual and the true perturbation, is uncorrelated with the advantage (Spearman 0.27) and does not separate the advantage units from the inversion units. A stronger measure, whether patching the residual reproduces the true output change, matches locality but no more (Spearman 0.59) and misses on the same cells: truncation has a large advantage yet low patch fidelity, and shuffling has high patch fidelity yet none. Four scalars have now failed to forecast the sign: no gate, map fit, locality, and survival in two forms. The advantage is a tail phenomenon, a false-clear rate at a matched flag quantile, and no global statistic of the representation or the patch that we have tried captures it. A forecaster, if one exists, is future work.

7 Consequence for monitor design

Three preregistered predictors of the consumer read’s sign have now failed, no gate, the equivariance map’s fit, and transform locality, and a fourth, perturbation survival, fails when tested post-hoc. The design consequence is the conservative one that this record actually supports. A consumer-metric equivariance gate cannot be certified safe from a calibration-time statistic of the representation, because the natural candidates do not forecast when the read inverts, and the inversion is large, up to 57 points of added false-clears. The defensible rule is to keep the raw metric as the default and to adopt the consumer read only for a transform family on which it has been validated empirically, on that family’s own held-out data, rather than extrapolated through any scalar. This is weaker than the gate we set out to find, and it is what the evidence licenses. The two robust findings stand alongside it: the paraphrase inversion is a measured warning against reading through the model under representation-relocating transforms, and the final-layer map collapse is a reason not to calibrate the monitor at the final layer under any norm.

8 Scope and limitations

Two model scales of one family, Social-Chemistry text, backtranslation, two paraphraser, and a synonym / shuffle / truncation string ladder, with the consumer read measured in x ’s context. The paraphrase inversion and the final-layer map collapse are robust across every run measured. The locality law is refuted on the transforms we tested and we make no claim that some other representation statistic could not forecast the sign, only that the three we sealed do not. We claim

nothing about monitors whose consumer is an external evaluator rather than the model’s own output, for which the read operator would have to be re-derived.

Reproducibility

Every number is read from committed, hashed records in the campaign repository: three sealed preregistrations, per-cell graded JSONs, post-mortems, and the locality gate with the checksum that reproduces the earlier families’ identity R^2 . Figure 1 and Table 1 regenerate from those records by script.

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